

OPEN RAIL

Jing-Zhang AI Commons • Urban Design Proposal (A3 Booklet)

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Open call: Centennial Jing-Zhang AI Innovation Belt open call

Iteration: v2.4 • 2026-08

Nature: Concept proposal / reference scheme; not a statutory plan or government conclusion

All areas and ratios in this booklet are recomputable from package GeoJSON in EPSG:4548. Boundaries are provisional; full recalculation is required upon official data release.

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01 Design Basis & Sources

The primary authority is the official pre-qualification announcement [source:OFFICIAL-ANNOUNCEMENT]; followed by the public site package [source:SITE-PACKAGE] and the agent taskbook [source:AGENT-TASKBOOK]; cultural materials serve narrative only [source:JINGZHANG-HERITAGE-PARK]. Source usability follows the public registry [source:SOURCE-REGISTRY].

Source	Level	Usage
Official announcement	official_public	Tasks, scopes, areas
Site package	official_public	Structured brief & rules
Agent taskbook	user_provided_cleared	Six tasks & boundary clause
Provisional boundaries	provisional_only	Interim generation; not redline
Heritage park public info	background_only	Narrative background

02 Three-Level Scope & Boundary

Level	Area	Extent
Research area	43.6 km ²	N 5th Ring / Jingzang Expwy / Xizhimenwai / Wanquanhe
Design area	11.4 km ²	1–2 km around heritage park (recomputed 11,412,825 sqm, +0.11%)
Key areas	368.4 ha	Zhongzhiyuan 192.1 / AI Origin 104.3 / Dazhongsi 72.0 ha

Official redlines are unpublished; all geometry derives from provisional_boundaries.geojson (provisional_rough) and is not for approval or precise area use.

03 Existing Conditions & Gap List

- The heritage corridor runs north–south but splits east–west links; six crossings have poor walking experience.
- Campus, community and park frontages are closed; innovation flows rely on arterial roads.
- Qing River and Xiaoyue River waterfronts lack systematic connection with the corridor.
- AI scenarios lack accessible, testable, displayable public-space carriers.

Data gaps (pending official data)

- Official precise boundary & key-area polygons
- Current regulatory controls (FAR, height, density)
- Existing buildings, property rights, municipal capacity

04 Research Area: Industry & Future-City Strategy

Concept： OPEN RAIL · Jing-Zhang AI Commons: the 9.7 km corridor as civic mainboard; rails as data & slow-mobility bus; stations as compute-community nodes; three areas as chips.

Three positionings form a timeline trio; five functions are anchored: full-stack to Zhongzhiyuan, ecosystem to Origin Community, scenarios to spine + Xiaoyuehe wing, vitality city to the slow network, governance voice to the testbed.

Six global cases distill transferable mechanisms of corridor-as-ecosystem, heritage-as-scenario, station-as-service [source:GLOBAL-AI-ECOSYSTEM-CASES].

Regional synergy (proposals, pending negotiation)

Partner	Exchangable resources	Interface & verification
Beiwei Community	Developer community, open-source branding, young talent	Wudaokou interface + honour-wall recognition; quarterly stats
Future Science City	Compute & energy experience, SOE R&D talent	Zhongzhiyuan compute access; joint-lab agreements
Huairou Science City	AI-for-Science demand, compliant data cooperation	Origin annual exhibition; shared filing tools
Beijing E-Town	Embodied-AI lines, robot supply chain	Dazhongsi front store; joint test reports
Jing-Jin-Ji region	Regional compute network, talent commuting & housing	Qinghe compute access; annual housing review

All synergies are conceptual proposals requiring partner agreement; no government commitments presumed.

05 Overall Design: Structure & Land Use

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Land-Use Structure: spine as axis, parcels as clusters

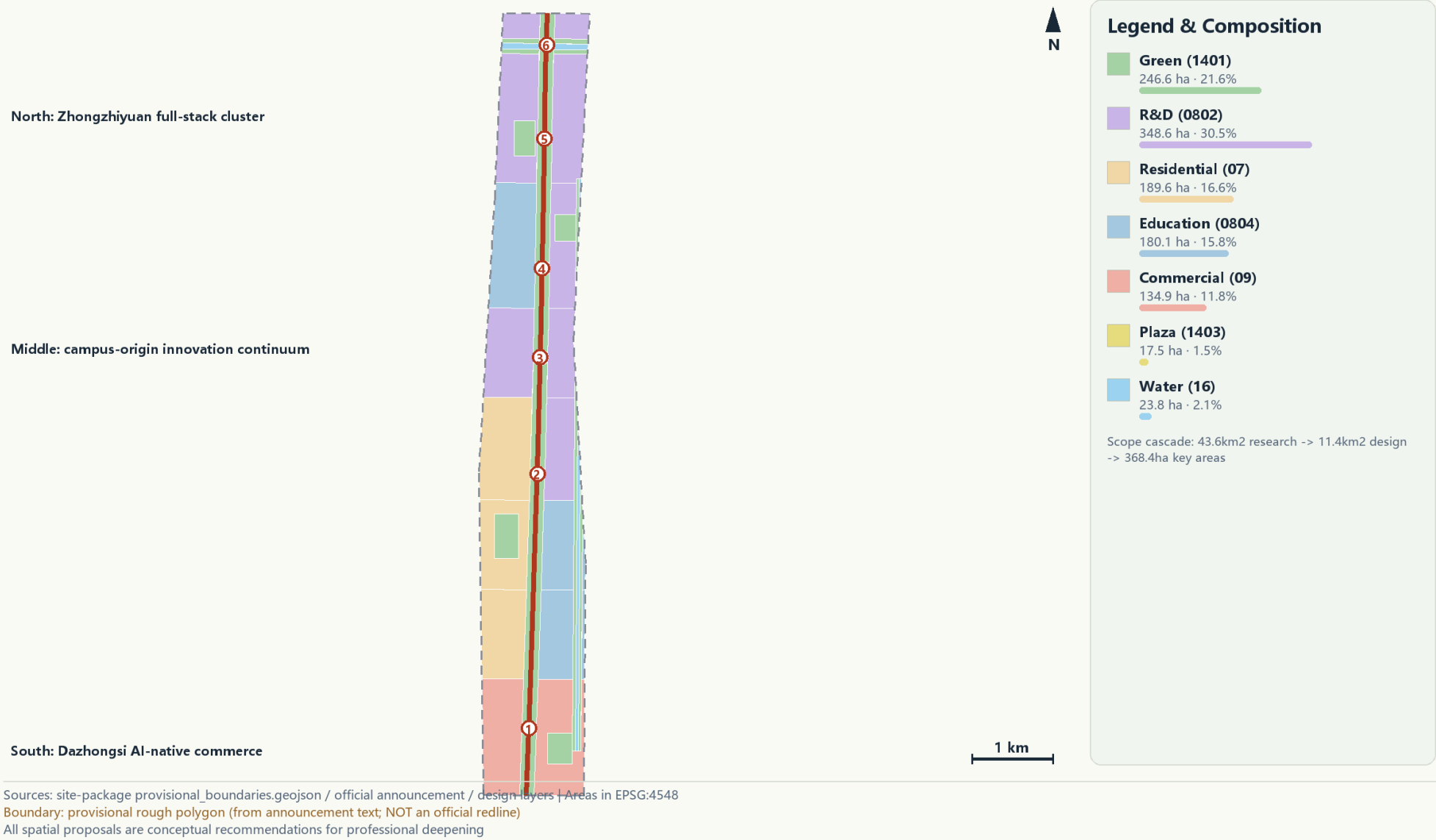


Fig.1 | Land-use structure (conceptual, provisional boundary)

Use	Area	Share
R&D 0802	348.6 ha	30.5%
Green 1401	246.6 ha	21.6%
Residential 07	189.6 ha	16.6%
Education 0804	180.1 ha	15.8%
Commercial 09	134.9 ha	11.8%
Water/plaza	41.4 ha	3.6%

Structure: 180 m corridor as spine, six halls as joints, three areas as chips, two wings as interfaces.

05b Overall Design: Mobility & Blue-Green

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Mobility x Blue-Green: stitched corridors, riverfront pearls

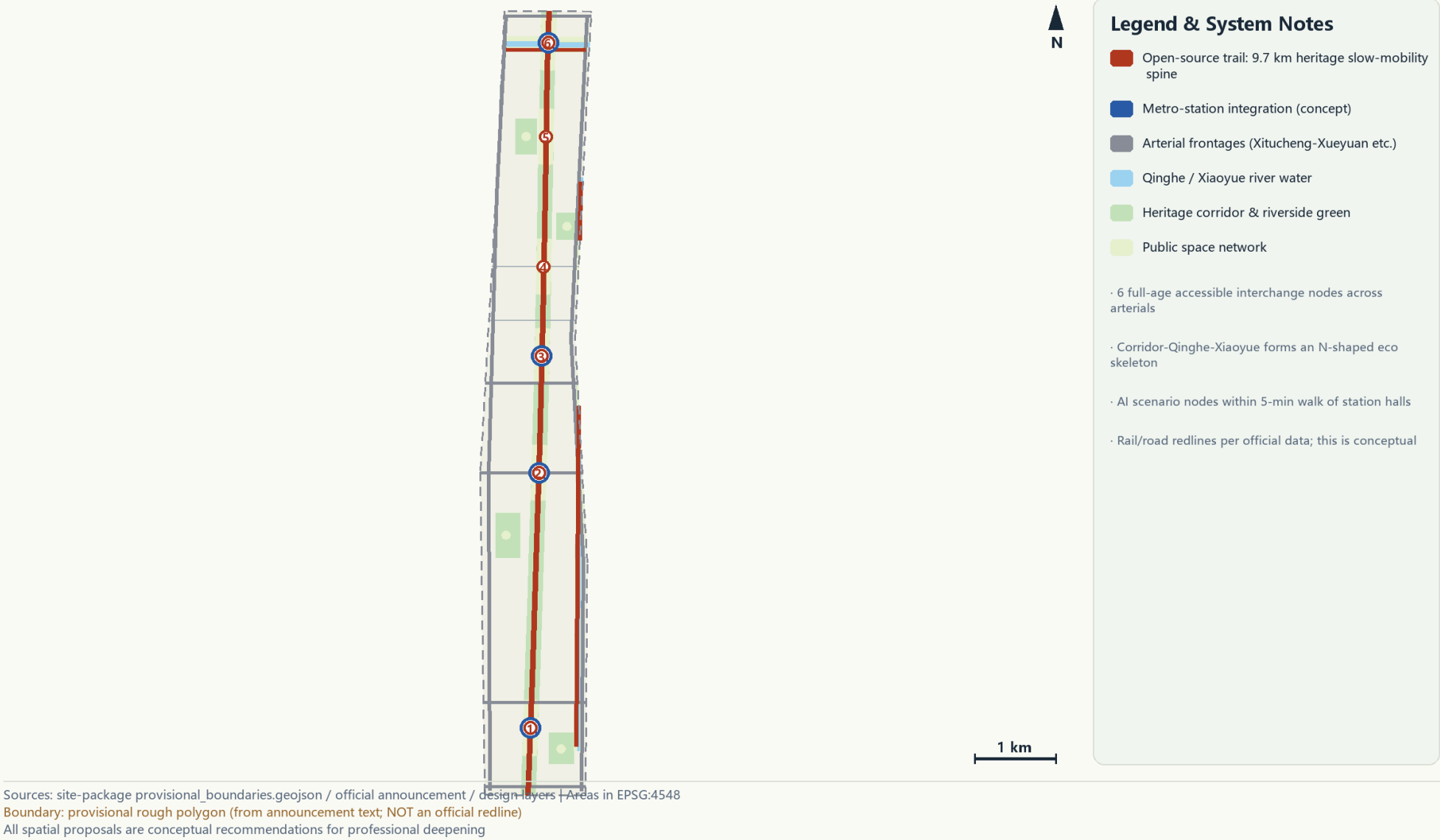


Fig.2 | Mobility and blue-green system (conceptual)

- 9.72 km trail runs north-south; six halls stitch east-west; metro integration is conceptual.
- Green ratio 21.61% and public-space ratio 10.56% (design-model, recomputable).
- Edge-compute nodes at six halls; distributed energy and sponge facilities along riverside green.

06 Key-Area Detailed Design

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Three Key Areas: positioning, anchors, risk conditions

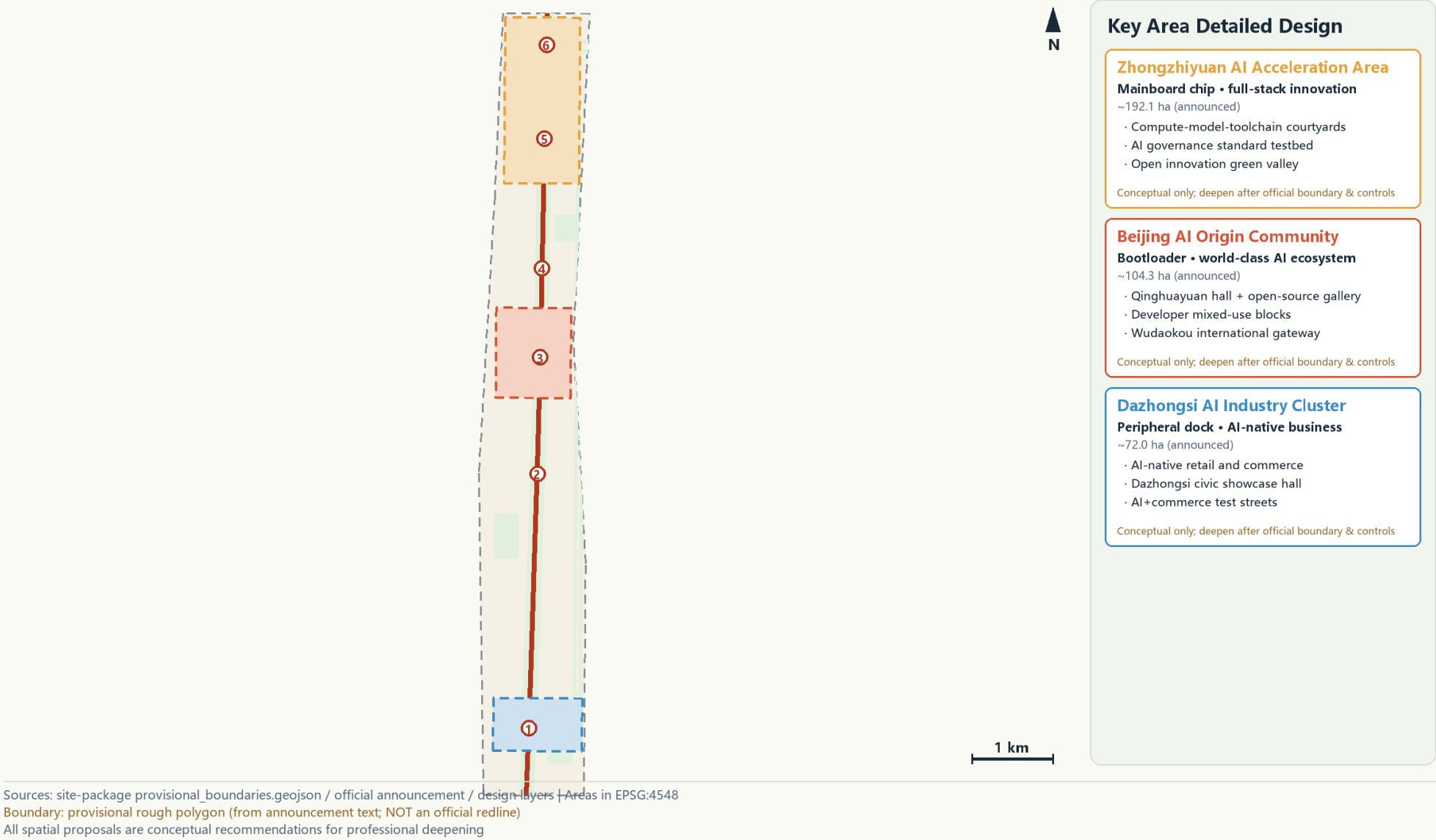


Fig.3 | Key areas and design tasks (conceptual)

Area	Role	Anchors
Zhongzhiyuan 192.1ha	Mainboard • full-stack	Compute-model-toolchain courtyards, governance testbed, green valley
AI Origin 104.3ha	Bootloader • ecosystem	Qinghuayuan gallery, developer blocks, Wudaokou gateway
Dazhongsi 72.0ha	Peripheral dock • AI-native commerce	AI-native retail, civic showcase, AI+commerce tests

06b Place Studies (1): Halls & Brand

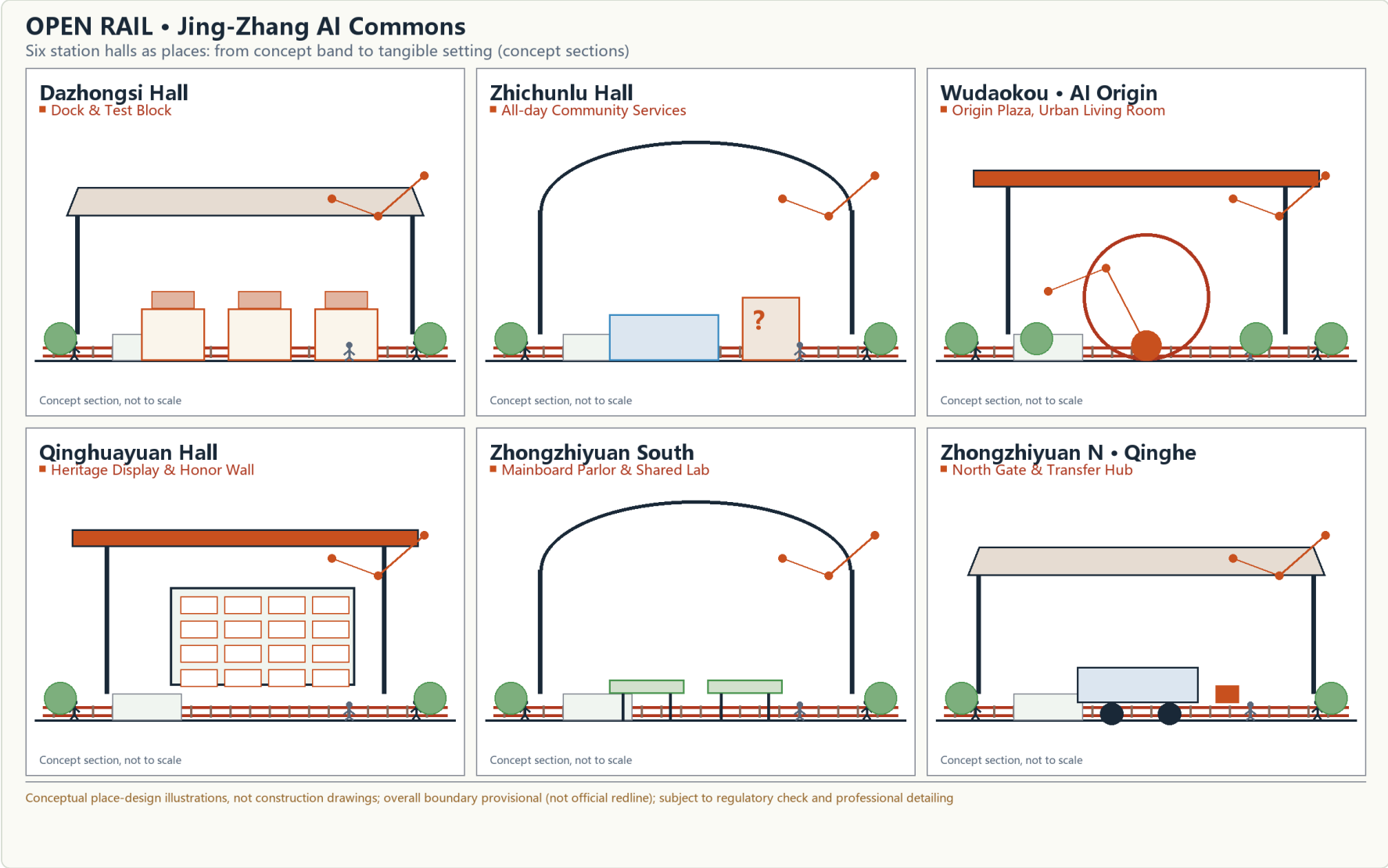


Fig.3b | Six halls as places (concept sections, not construction drawings)

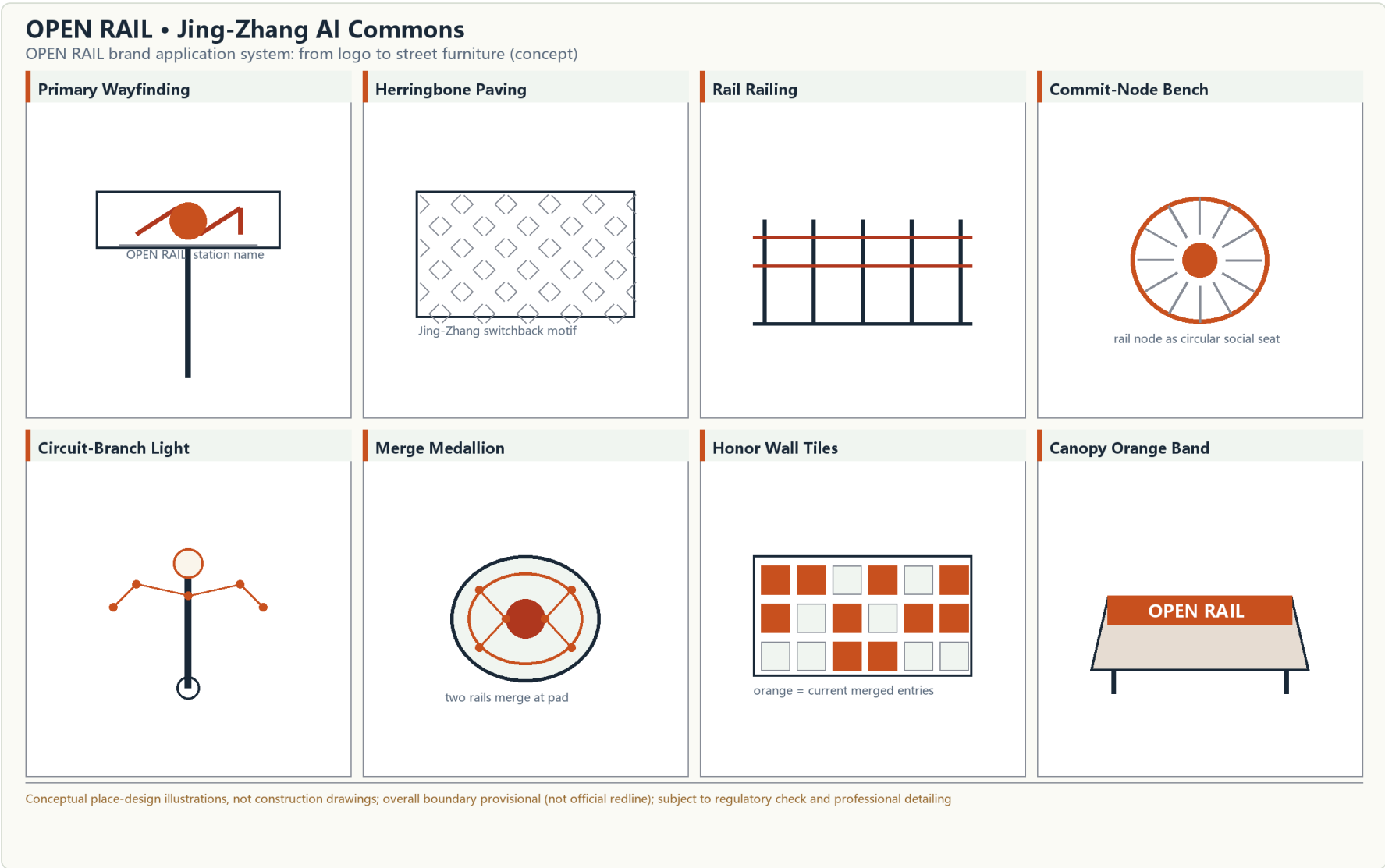


Fig.3c | OPEN RAIL brand application system

06b Place Studies (2): Commit Plaza Node

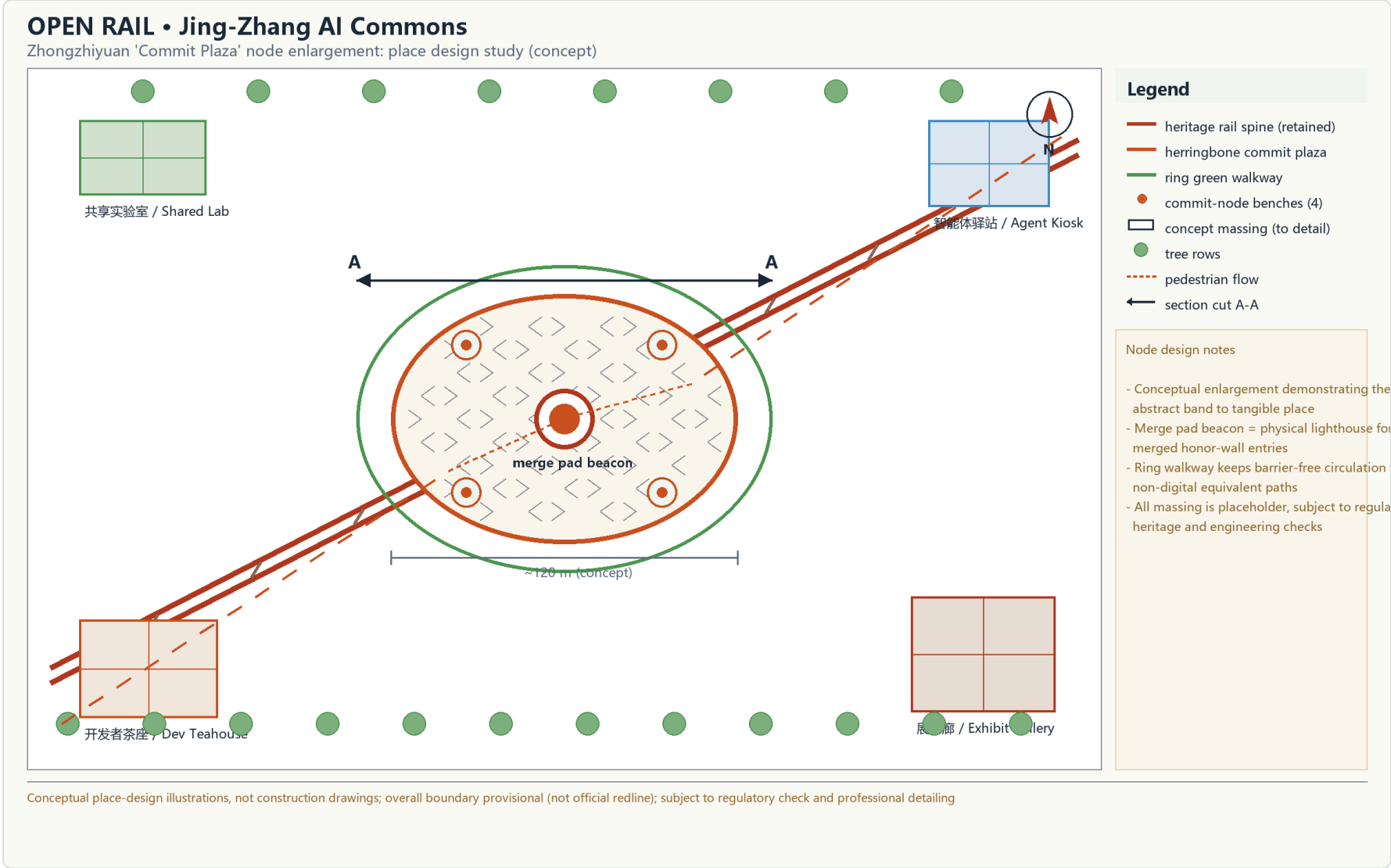


Fig.3d | Commit Plaza node enlargement (placeholder massing)

From abstract band to tangible place: hall sections, branded furniture and a node enlargement demonstrate the direction. All are concept studies subject to regulatory, heritage and professional checks.

06c Streetfront & Construction Studies

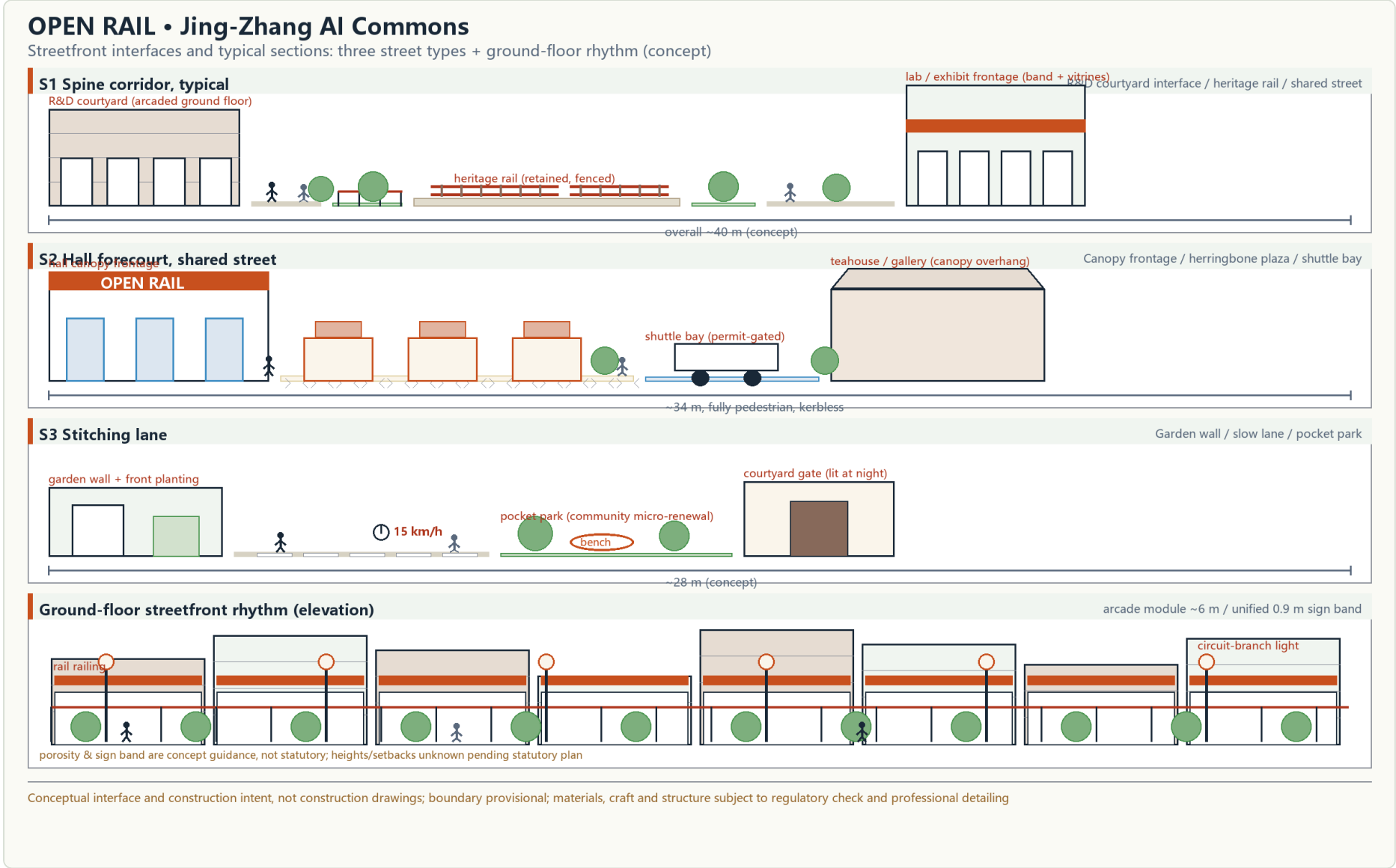


Fig.3e | Streetfront interfaces & typical sections (S1 corridor / S2 forecourt / S3 lane + ground-floor rhythm; concept)

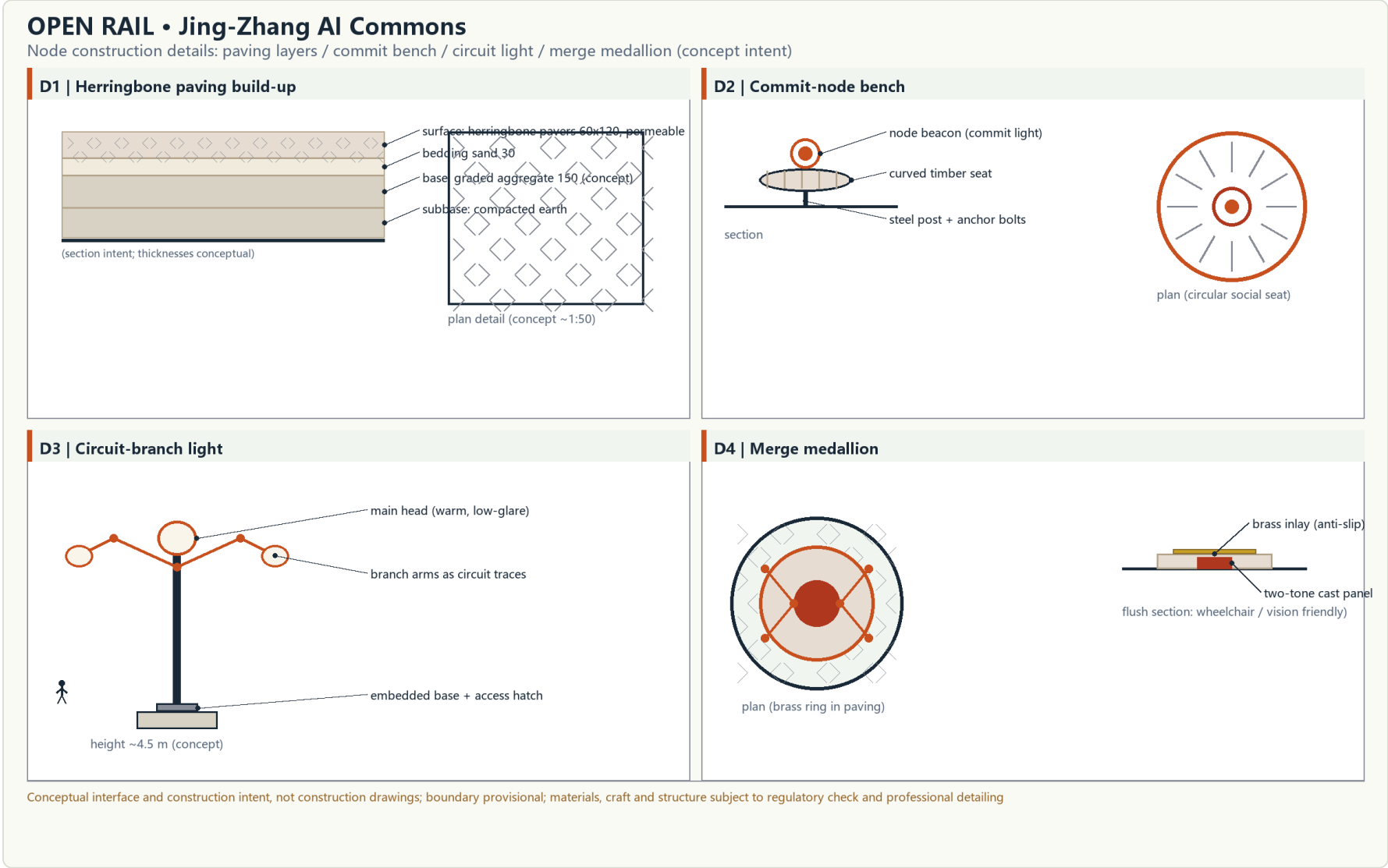


Fig.3f | Node construction details: paving / bench / light / medallion (intent, not construction drawings)

Heights/setbacks unknown pending statutory plan; materials and structure follow professional detailing and accessibility review.

07 AI+ Scenarios & Data Governance

13 scenario cards all carry data boundaries and human review; 3 test scenarios require authority permission:

AI+walkability

AIcultural guide

AIhealth nav

enterprise copilot

safety review

test: robot delivery

AI+open class

AI+retail guide

edge-compute test

smart park O&M

test: autonomous shuttle

test: governance sandbox

AI accessibility companion (non-digital equivalents)

Six personas

Researchers / developers / students & faculty / residents / visitors / older adults & accessibility needs — each mapped to space supply, scenarios and operations.

AI pilgrimage landmarks (4 concepts)

Developer Promenade

Agent Honour Wall

Open-Source Gallery

AI Origin Monument

Privacy & governance

Personal data stays on device, authorised use, human review; no excessive surveillance; tests are segment-limited, speed-limited, takeover-ready.
Scenario governance: component-level adoption of community open standards — Switchback Protocol v1.0 and SEB v0.5.0 node schema; non-official methods, not statutory bases.

08 Renewal Projects & Phasing

Phase	Scope	Key projects
Near 2026–29	S. spine + Origin + Dazhongsi	1,2,3,5
Mid 2029–32	N. corridor + Zhongzhiyuan S	1,4,6,7
Far 2032–35	Zhongzhiyuan N + Qinghe	7,8

Projects: 1 spine link; 2 six halls; 3 Qinghuayuan gallery; 4 green valley; 5 Dazhongsi renewal; 6 housing + pocket parks; 7 riverside link; 8 edge-compute demo. Bodies, funding and approvals follow formal decisions.

Annual events & operations (concept)

Spring conference / summer test season / autumn culture season / winter agent annual; GitHub PR/Issue as standing co-creation routine.

Funding typology & entity paths (proposals)

Type	Applies to	Entity path	Boundary
City/district funds	1&7 public works	District projects, statutory	Per formal approval
Market investment	5 blocks, courtyards	Concession/co-development	Property negotiation first
Industry funds	Scenarios & incubation	Open scenarios + matchmaking	No promises presumed
Foundations	Cultural content	Charter, disclosed	Heritage review first
Community micro-funds	6 stitching nodes	Participatory budgeting	Consent threshold
Operations reinvest	Events & operations	Disclosed under review	Audit scope TBD

Baseline protocols (pre-launch)

P1 accessibility coverage (quarterly) / P2 equivalent paths / P3 response median / P4 walkability breaks / P5 third-party audit; failures retained; unmeasured stays pending-baseline.

Sequencing & cost-benefit methodology (proposal, trigger-gated)

Step 1 data baseline (on official geometry/statutory release) / 2 priority scoring (weighted, negotiable) / 3 cost lines (no estimates) / 4 benefit lines (P1-P5 + proxies) / 5 sensitivity (robust intervals). No ranking or benefit figures before all triggers; conclusions jointly reviewed and rolled with the 90-day review.

09 Metrics, Recalculation & Compliance

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Metrics & Evidence: recomputable from package geometry

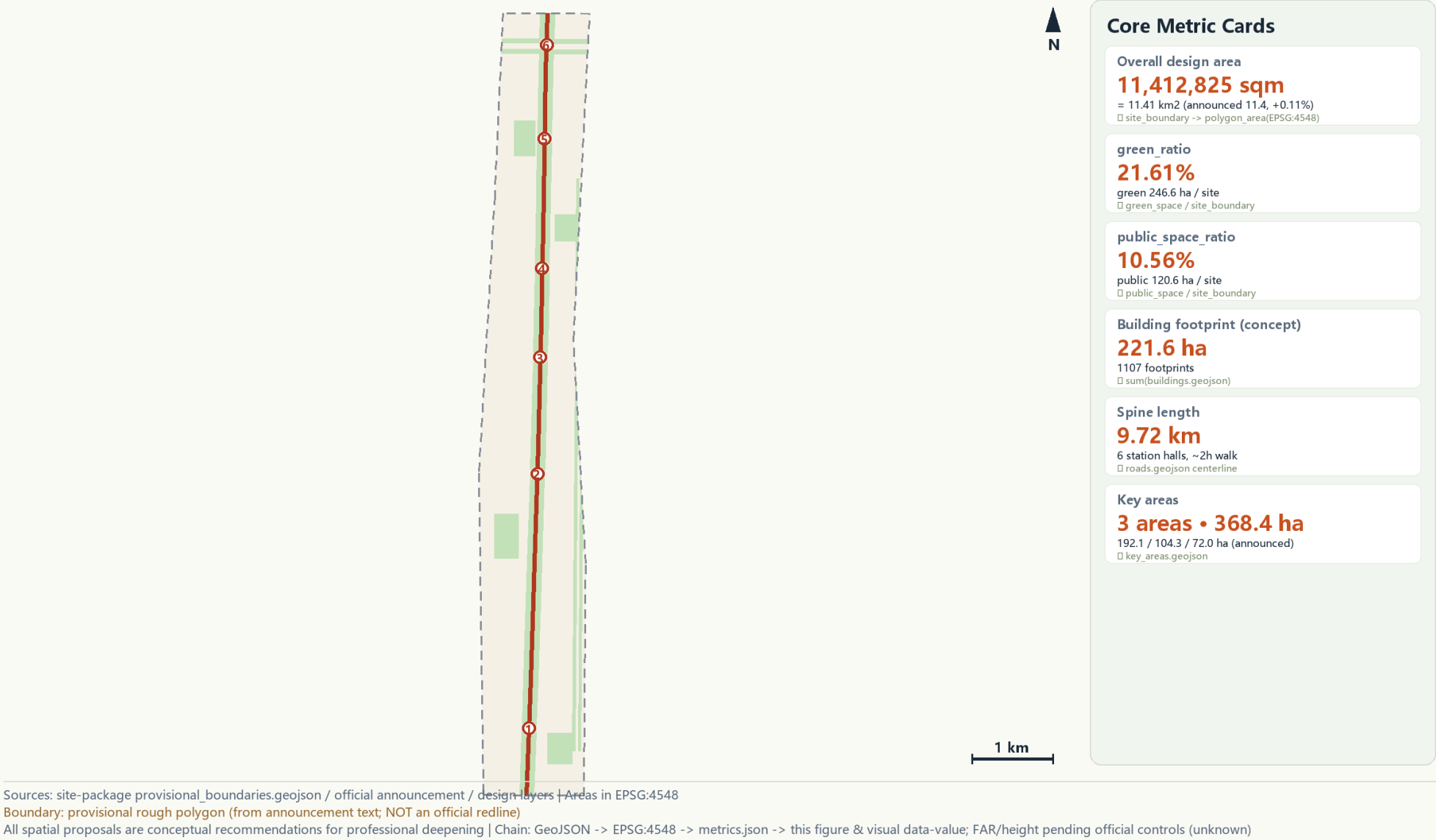


Fig.4 | Metrics recalculation and evidence chain

11,412,825 m ² design area (recomputed)	21.61% green ratio	10.56% public space
221.6 ha footprints (concept)	9.72 km spine length	368.4 ha key areas

FAR/height: unknown pending controls. Compliance matrix covers all 23 requirements (1.3/1.4/1.5 + agent.1-6); standard matrix covers mandatory standards; all 15 depth items complete.

10 Risk, Copyright & Missing Data

- Provisional boundary: full recalculation upon official data (A-BOUNDARY-002).
- Regulatory gap: intensity/height/demolition pending (A-CONTROLS-001).
- Heritage: Qinghuayuan content is conceptual, subject to relics protection.
- Implementation: phasing and projects are advisory, not commitments.

Generated by an AI agent from public/cleared sources; figures are original programmatic renderings. See report/copyright_statement.md.

Bilingual

This booklet has an English counterpart; proposal.md (zh) and proposal.en.md (en) mirror chapters, claims, metrics and figures.